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BITS Pilani
Pilani Campus

Wood & Timber

Introduction



- **Wood:** An organic matter, hard and fibrous substance obtained from trees
- **Timber:** The wood which is suitable as a building material or fit for Engineering construction
- **Converted timber/Lumber:** Logs of timber sawn into planks, posts, etc
- Wood as a building material is classified as:
 1. Natural: Used in natural form as Timber, Lumber etc.
 2. Man-made: Composite wood materials in which natural wood is just a basic ingredient of a matrix or a laminate. Ex: Plywood, fiber boards, impregnated wood, chipboard etc.,



Introduction



Advantages of wood as building material

- Easily available?
- Easy to transport and handle
- More thermal insulation
- Good sound absorption and prevents echo
- High electrical resistance
- Earthquake resistant construction as wood is a good shocks absorber
- Easily worked, repairs and alterations are easy
- Used as decorative material in aesthetic aspects

Wood as building material



Doors



Partition walls



Ceiling and window frame



False ceiling and aesthetics



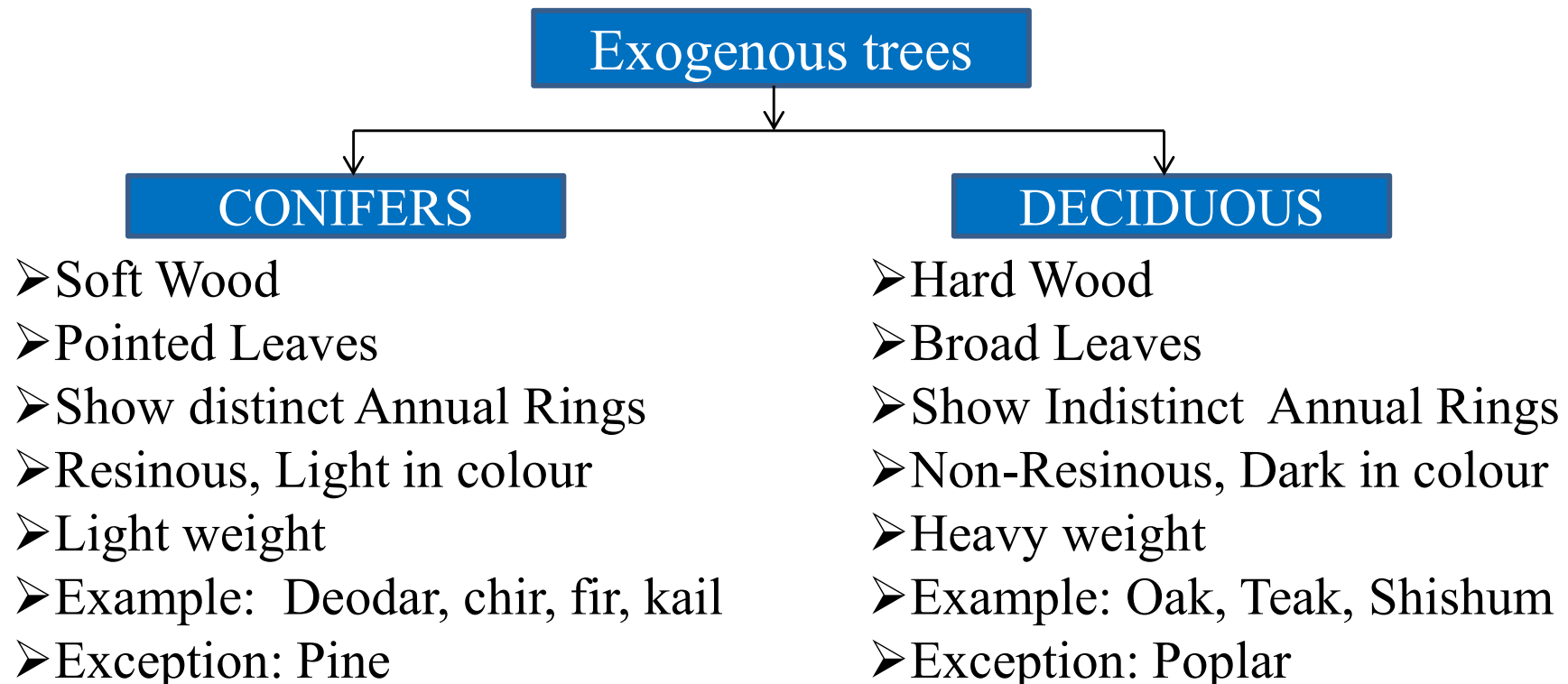
Roof Truss

Classification of trees



Based on Mode of growth

- Endogenous Trees: Trees which grow **endwards** in **longitudinal** fibrous mass. Ex. Palm & Bamboo
- Exogenous Trees: Trees which grow **outwards** across **horizontal** section of stem. These are used for making structural elements.



Soft Wood & Hard Wood



Property	Soft Wood	Hard Wood
Colour	Lighter	Darker
Growth	Faster	Slower
Weight	Lighter	Heavier
Density	Low	High
Annual Rings	Distinct	Indistinct
Heart wood & Sap wood	Cannot be distinguished	Can be distinguished
Strength	Strong along the grains	Strong along and across the grains
Conversion	Easy	Difficult
Resinous material	Exists in pores	Do not exist

Growth of tree

SPRING SEASON

Tree roots suck sap (food)

Sap reaches Branches and leaves

Moisture in Sap gets evaporated

In daytime, Sap absorbs carbon & becomes denser

AUTUMN SEASON

This Sap forms a layer below bark called CAMBIUM

Forms every year and add as a layer of wood – ANNUAL RING



Growth of tree...contd

- These **annual rings** gives significant data such as **age**, **rapidity** and the **uniformity** of its growth.

The strength of wood also depends upon the grain type

- Rapid growing trees – Wider annual rings – **Coarse grained** wood
- Slow growing trees – Narrow annual rings – **Fine grained** wood
- Wood elements running longitudinal or parallel to axis or pith – **Straight grained** wood
- If not running parallel - **Cross grained** wood

Macrostructure of timber



The various components in the cross section are explained below

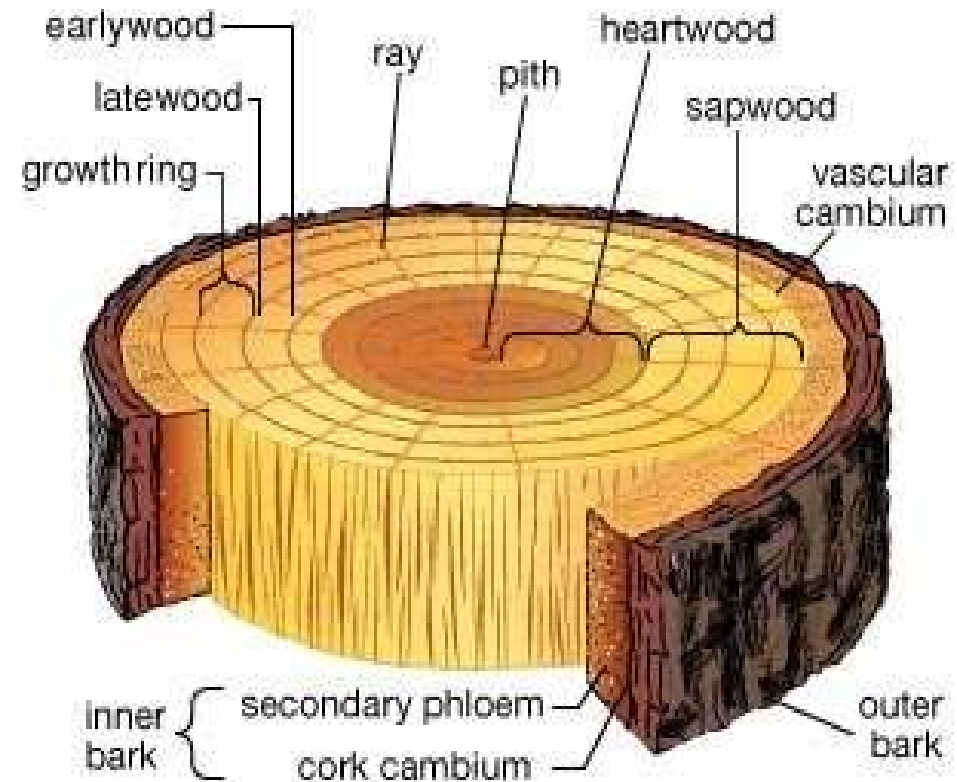
- Each ring indicates 1 year growth, called as annual ring.

Pith:

- The inner central portion
- Consists of cellular tissues and nourishes the plant at young age.
- It dies and decays upon ageing

Heart Wood:

- Dark in colour
- Indicates the dead portion of tree.
- Imparts rigidity and durability.



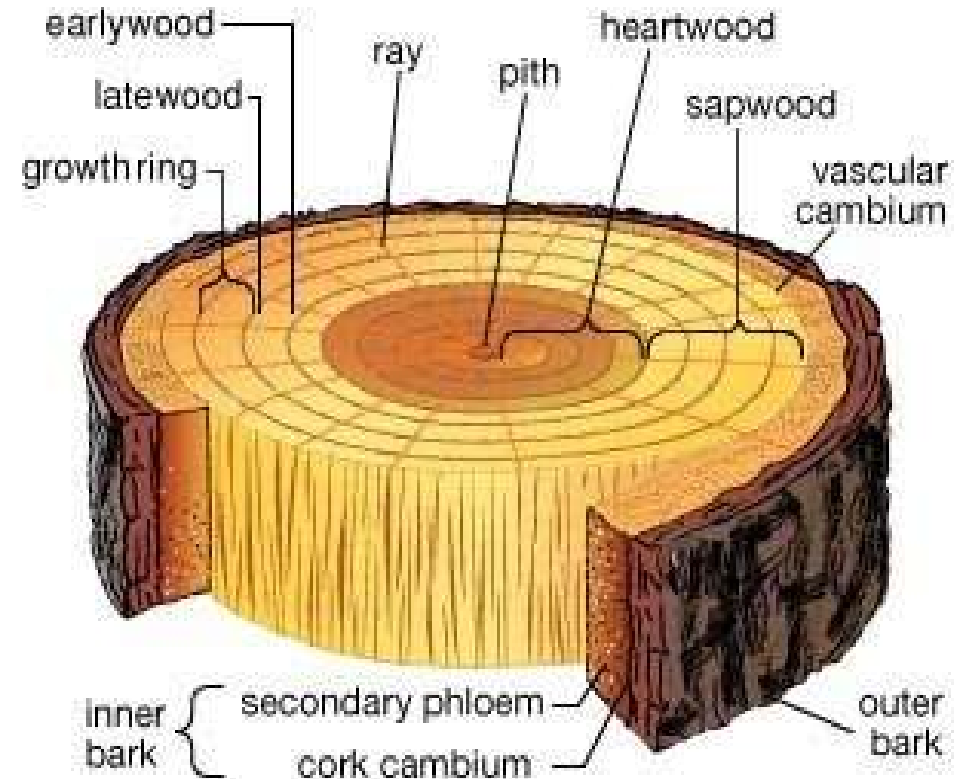
Structure of well grown timber from trunk of the exogenous tree

Macrostructure of timber



Sap Wood:

- Outer annual rings portion between heart wood and cambium layer containing sap and indicates growth
- Light weight and light colour
- Annual rings less sharper than that of heart wood
- Conducts moisture from roots and stores food



**Macroscopic Structure of well
grown timber from trunk of the
exogenous tree**

10

Macrostructure of timber

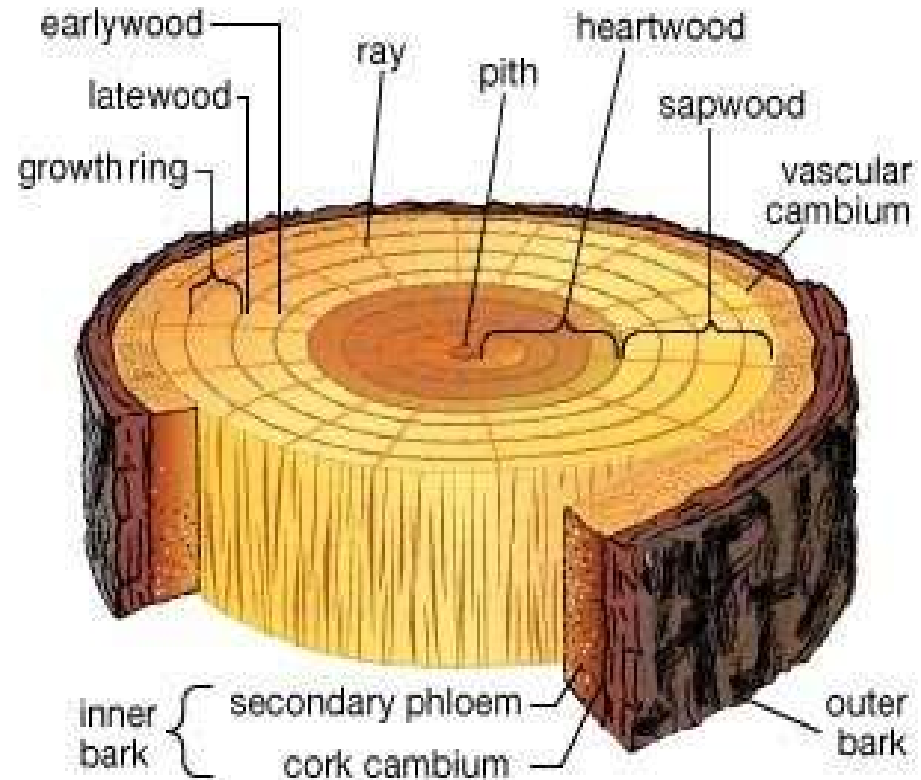


Cambium Layer:

- Thin layer of sap between Sapwood and inner bark
- Indicates that sap is not yet converted into Sapwood
- If the cambium layer is exposed by removing the bark, the cells cease to be active and results in death of tree.

Bark:

- Protects the tree against mechanical damage



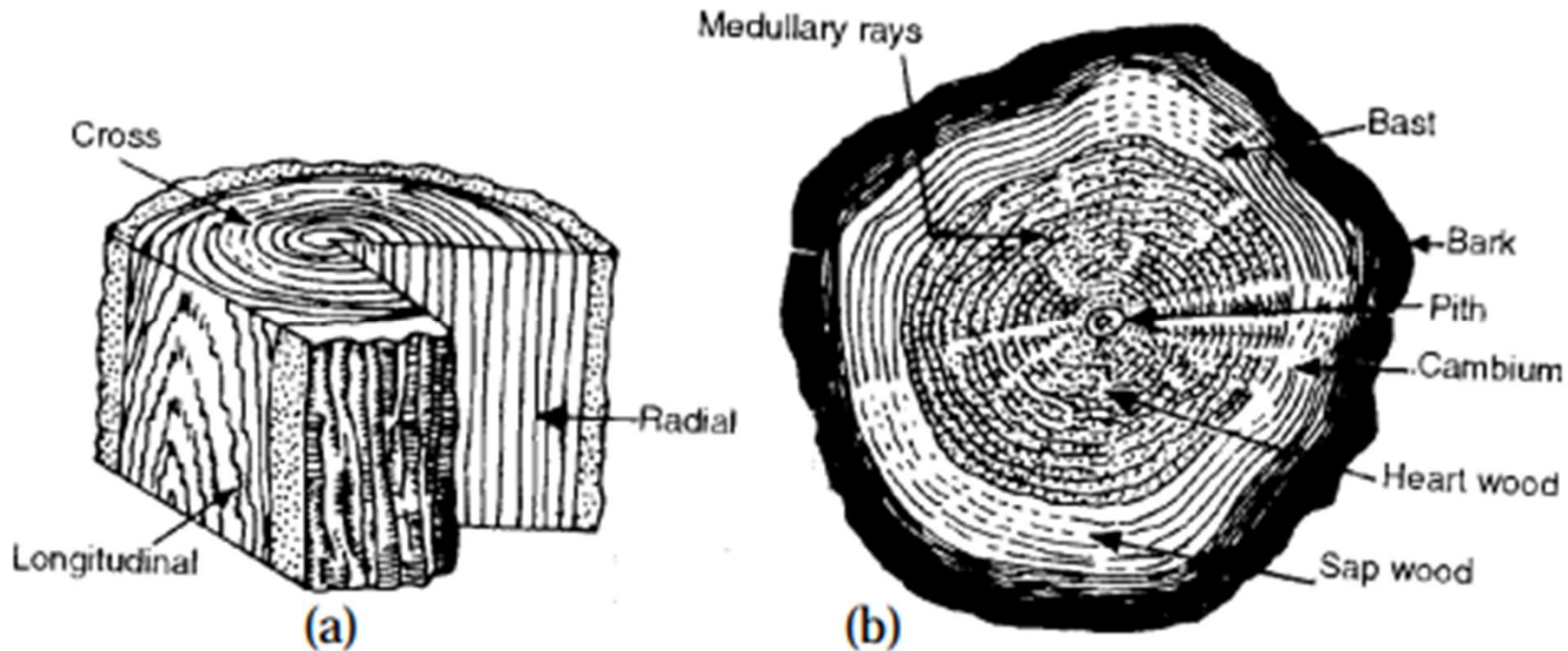
Macroscopic Structure of well grown timber from trunk of the exogenous tree

Macrostructure of timber



Medullar Rays:

- Thin radial fibres extending from pith towards the bark
- Carry sap from outside to inner parts of tree and nourish it
- Keep the annual rings tightly gripped together



Characteristics of Good Timber



Basic characteristics required are strength, durability and appearance

Narrow annual rings

Compact medullar rays

Dark color

Uniform texture

Sweet smell and a shining fresh cut surface

When struck sonorous sound is produced

Free from the defects in timber

Heavy weight

Seasoning of timber

- Fresh timber contains about 30 to 40% sap or moisture
- Seasoning is process of reducing the moisture content (drying) of timber to prevent it from possible fermentation and making it suitable for use
- Rapid seasoning after removal of bark should be avoided as it causes case hardening and resists penetration of preservative

Reduced shrinkage and warping after placement in structure

Increase in Strength, durability and workability

Reduced tendency to split and decay

Reduced weight

Easy to polish/paint

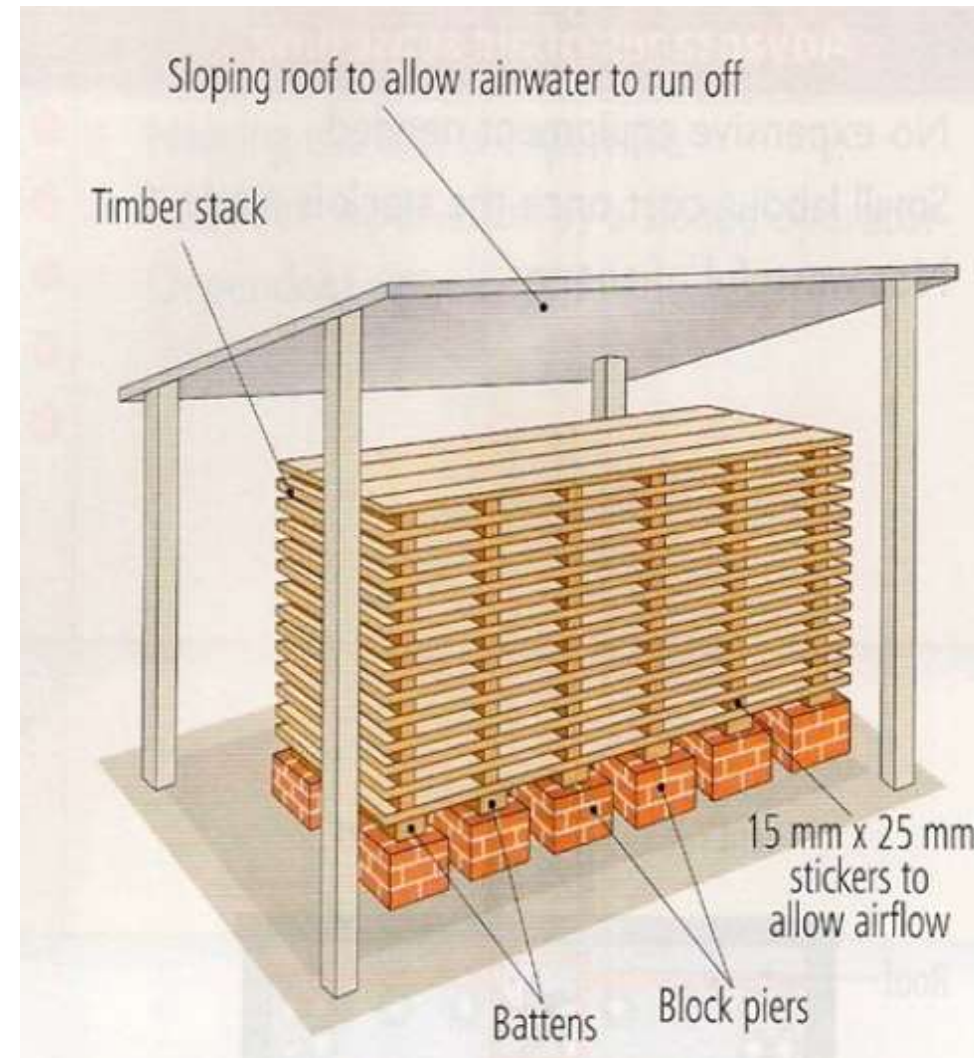
Methods of seasoning



Timber can be seasoned naturally or artificially

Natural or Air Seasoning:

- Timber is allowed to dry by the action of wind, air and sun.
- In this method, the timber logs are placed in cross wind direction in alternate layers (with some spacing) to allow free air circulation.
- Stacks covered to prevent rain
- Ends protected from sun by painting or metal cleats to prevent excess drying out
- Used extensively in drying ties and large structural members.



Methods of seasoning...contd

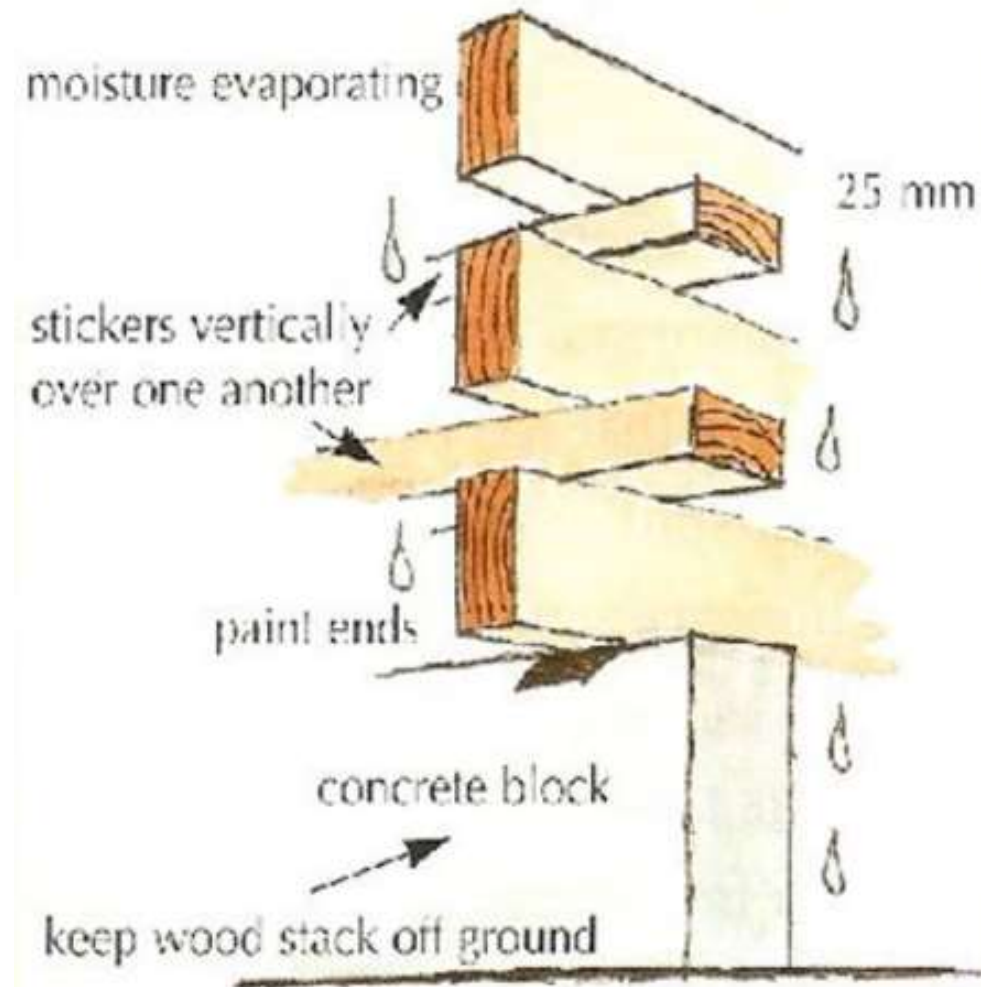
Natural Seasoning

Advantages

- No expensive equipment required
- No energy wastage
- Small labour required once the stack is made

Disadvantages

- Slow drying rate
- Large space area required
- Dependant on weather & rate of seasoning not under control
- Only seasons to a moisture content of 12-18%



Methods of seasoning...contd



ARTIFICIAL SEASONING:

- It is adopted to make drying more uniform and even under controlled conditions
- It reduces time of seasoning
- Shrinkage, cracking and warping are minimized
- The wood with desired moisture content can be obtained

Methods of seasoning...contd



Artificial Seasoning: The prevalent methods are as follows:

1. Water Seasoning

- Wood placed in running stream of water with their larger end pointing upstream. The sap, sugar and leach present would be replaced by water.
- Wood is then placed out in air to dry. Though quick process, but elastic properties and strength of wood is reduced.

2. Another artificial way of seasoning is to expose the wood to the action of **steam spray** which is very quick but expensive process.

Methods of seasoning...contd

3. Kiln Seasoning: Time required for seasoning is 3 – 12 days

❑ Advantages

- Quicker due to higher temperature, ventilation and air circulation
- Achieve low moisture content
- Defects associated with drying can be controlled
- Allows more precise rate of drying for various timber species

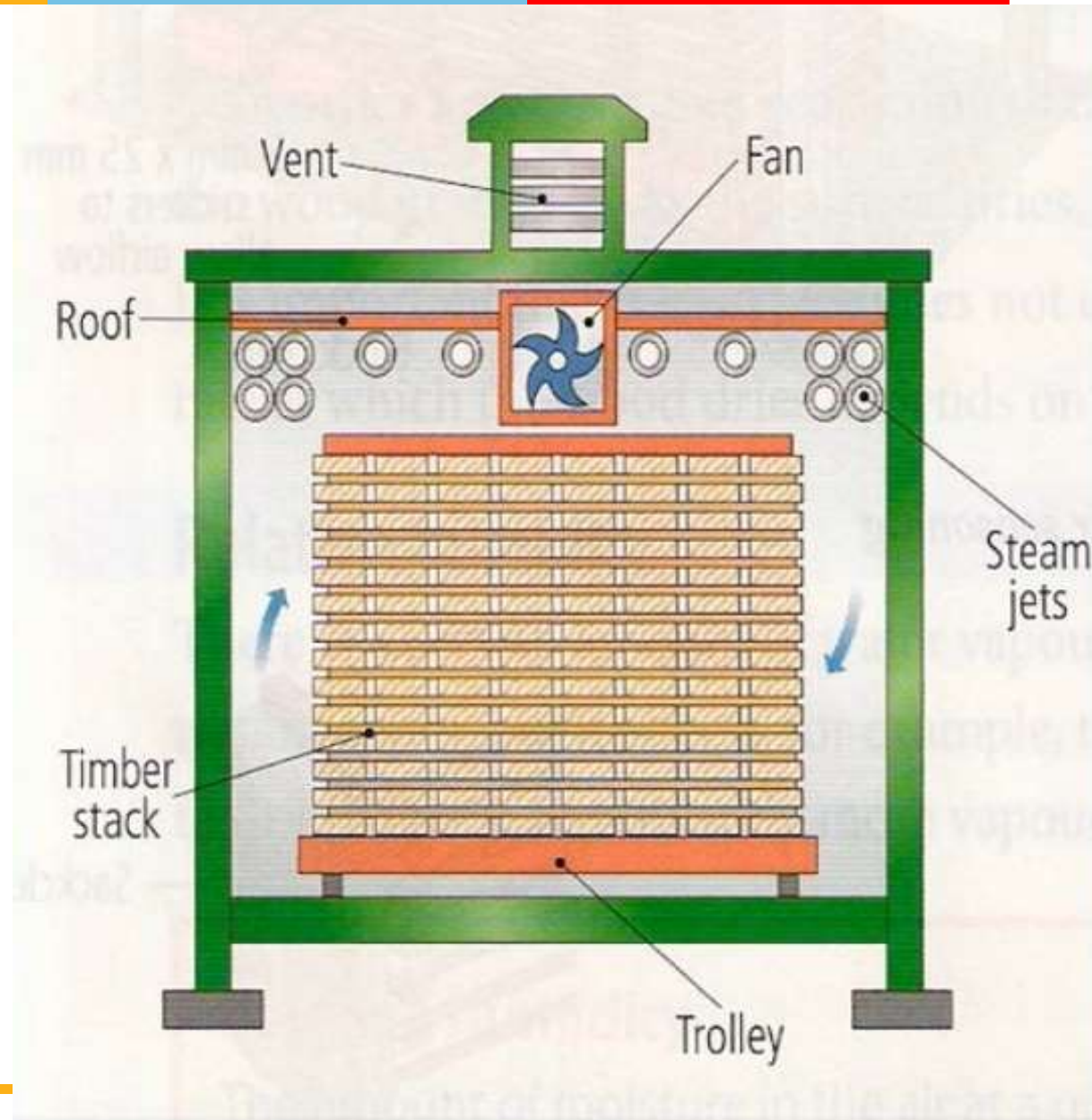
❑ Disadvantages

- Expensive
- Requires supervision of a skilled labour
- Uses immense energy

Methods of seasoning...contd



Kiln Seasoning



Methods of seasoning...contd



4. Chemical or Salt Seasoning:

- In this process, outer layer of timber treated with salt solution or chemical
- Later, it is treated by air seasoning.
- The interior of timber tends to dry up as the salt solution tends to absorb water.
- It results into reduction of internal stresses which induced during drying which are responsible for defects.
- Hence, a chemically treated timber has less defects.



Methods of seasoning...contd



5. Electric Seasoning:

- Electric current passed through timber logs.
- Being bad conductor, current flow is resisted due to which heat is generated, resulting in drying.
- Time required for seasoning is 5 to 8 hours.
- **Drawback:** Wood may split



Defects in Timber



➤ **DEFECT**: It is an irregularity or abnormality occurring in or on wood during its growing period, conversion or seasoning process which is responsible for its

1. Strength reduction
2. Lowering in Durability
3. Lowering of Utility
4. Poor Appearance
5. Decay

➤ It is caused due to

- ✓ **NATURAL FORCES**
- ✓ **INSECTS**
- ✓ **FUNGI**
- ✓ **CONVERSION**
- ✓ **SEASONING**



Defects in timber...contd



I. Defects due to Natural Forces

1. Chemical Stains:

- The wood gets discolored due to some chemical action within it or by some external agency.



2. Rind Galls

- “Rind” means Bark & “Galls” means Abnormal growth.
- Peculiar curved swelling found on the tree body are Rind Galls.
- Generally grown where branches are improperly cut off and removed.
- Timber in this part is Very Weak and Not Durable.



Defects in timber...contd



I. Defects due to Natural Forces

3. Knots

- These are bases of branches or limbs which are broken or cut off the tree.
- This portion forms dark hard rings after receiving nourishment from the stem for a long time which are known as Knots.
- As continuity of wood fibres is broken, they form a source of weakness
- Classified on the basis of size, form, quality and occurrence.

Based on **Size**:

Pin Knot < 12mm ; Small Knot 12 – 20mm

Medium Knot 20 – 40mm ; Large Knot > 40mm



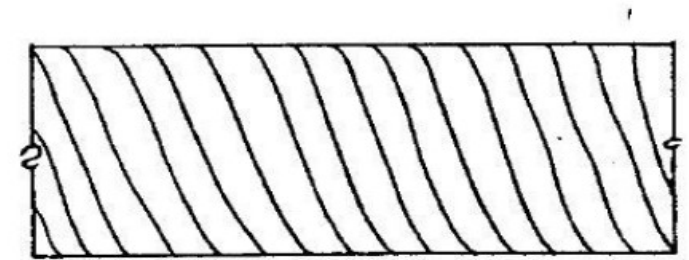
Defects in timber...contd



I. Defects due to Natural Forces

4. Twisted Fibres

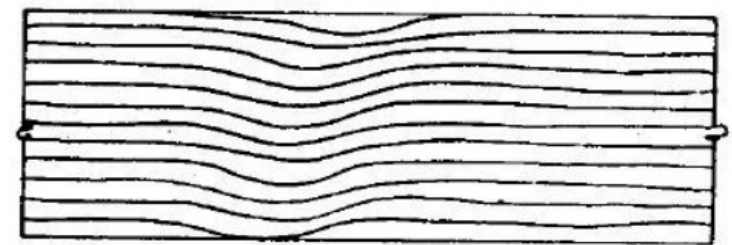
- Caused by twisting of young trees by fast blowing wind



Twisted fibres

5. Upsets

- Wood fibres which are injured by crushing or compression.



Upset

Defects in timber...contd



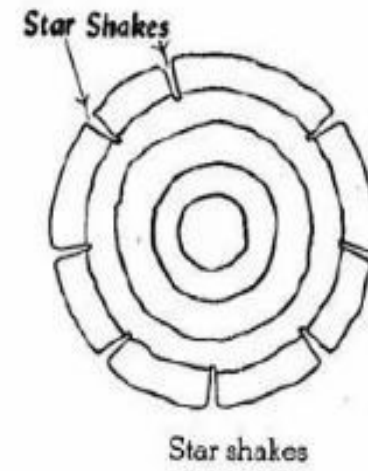
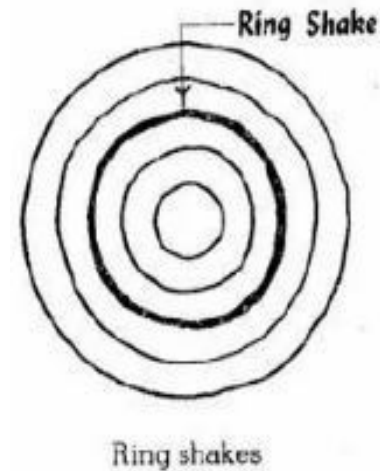
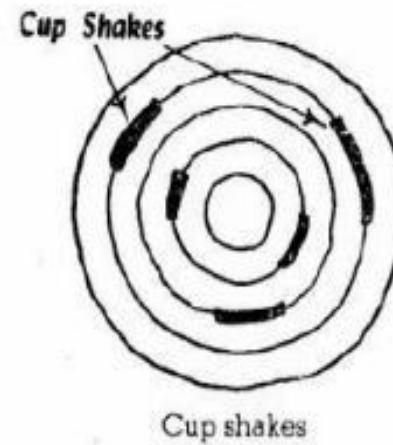
I. Defects due to Natural Forces

6. Shakes

- They are longitudinal separations in wood between the annual rings.
- This separation makes the wood undesirable when appearance is important
- Reduces allowable shear strength

Types of Shakes

1. Star Shakes
2. Cup Shakes
3. Heart Shakes
4. Ring Shakes



Defects in timber...contd



II. Defects due to Fungi

1. Blue Stain

- Sap of wood is stained by bluish colour by the action of certain type of fungi



2. Dry Rot

- Decomposition of felled timber due to fungal action . Also known as **Brown Rot**.
- Reduces fibres to fine powder and loose strength
- Occurs when imperfectly seasoned
- Placed in confined atmosphere with no air access
- Remedy: Cut away affect part and paint the remaining part



Defects in timber...contd



II. Defects due to Fungi

3. White Rot

- Fungal attack on lignin of wood
- Causes appearance of white mass having cellulose compounds



4. Heart Rot

- Formed when branch has come out of the tree
- The heart wood is exposed to the attack of atmospheric agents & tree becomes weak
- It gives Hollow sound when struck with hammer



Defects in timber...contd



II. Defects due to Fungi

5. Wet Rot

- Decomposition takes place due to alternate wetting and drying conditions
- Setup by the access of water through wounds of bark and cause decomposition of sap wood and fibres
- May occur when timber is seasoned by exposing it to moisture
- Remedy: Well seasoned timber is used with preservatives and paints



Defects in timber...contd



II. Defects due to Fungi

Control of Fungal attack

- Timber should not be subjected to **alternate wet and dry**.
- If unavoidable, proper preservative treatment should be made.
- Felled trees should air dried as soon as possible & Sawn timber should be kiln-seasoned properly in accordance with good air-seasoning practice.
- Protected from rain and other sources of moisture.
- Ensure adequate ventilation
- **Timber for Structural use should not contain Sap wood as more susceptible to fungal attack**

Defects in timber...contd



III. Defects due to Insects

1. Termites

- Lives in colony and very fast in eating away the wood from the core of cross-section
- Make tunnels in different direction and usually not disturb the outer shell
- The attacked timber piece may look sound until it completely fails
- Timbers like Teak, Sal etc., resist the action of termites



Defects in timber...contd



III. Defects due to Insects

2. Beetles

- Cause rapid decay of timber by **converting it into fine powder**.
- Tunnel formation by Larvae of beetle
- The attacked timber piece may look sound until it completely fails

3. Carpenter Ant

- They do not eat wood and **tunnel it for habitation**.
- Attack slightly rotted or water softened wood and continue if wood appears to be sound.

Defects in timber...contd



III. Defects due to Insects

Control of Insects

- Tunnels made are **penetrated with toxic elements** to destroy them
- Large scale fumigation carried out using hydrocyanic acid gas but not recommended as highly poisonous.
- **Alternative:** Turpentine + Small dose of orthodichlorobenzene
- Its vapour deadly to insects and not poisonous to human beings

Defects in timber...contd



IV. Defects due to Conversion

- **Conversion** – A process where felled tree is converted into marketable sizes of timber.
- The defects arise due to unsound practice in milling or attempts to economise during conversion of timber. Some defects are as follows:

Chip Mark: It is indicated by marks or signs placed on finished surface of timber.



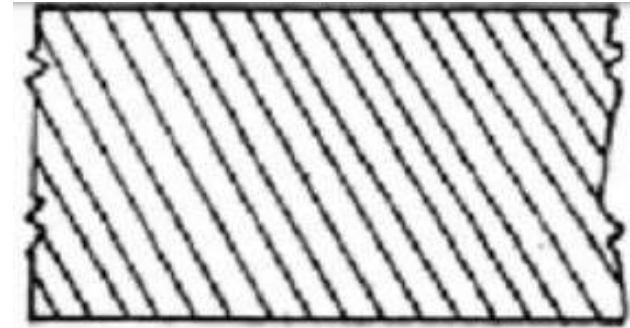
Wane: The presence of bark or the absence of wood on the corners or along the length of a piece of lumber.

Defects in timber...contd



IV. Defects due to Conversion

- **Diagonal Grain:** The defect formed due to improper sawing of timber.
- Indicated by diagonal marks on straight grained surface of timber.



Torn Grain: A defect caused when a small depression is formed on a finished surface of timber by falling a timber or so.

Defects in timber...contd



V. Defects due to Seasoning

➤ During Seasoning process, the wood may become warped or distorted due to change in MC. Some defects are as follows:

Bow: This defect is indicated by curvature formed in direction of length of timber.

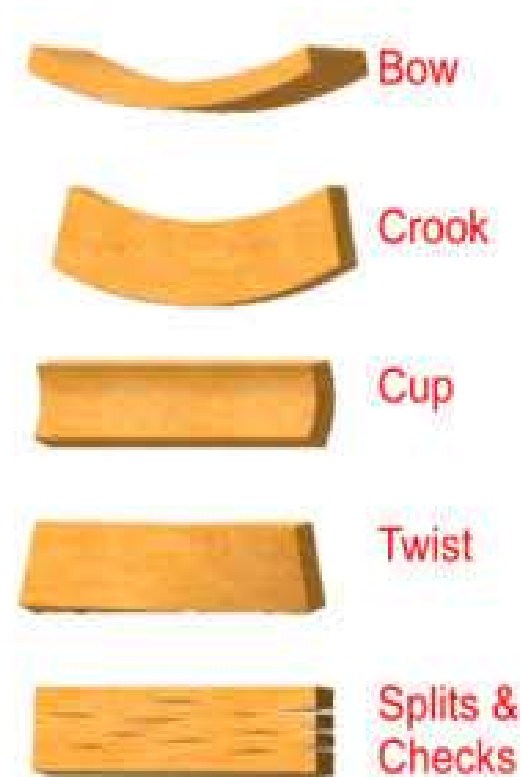
Cup: This defect is indicated by curvature formed in transverse direction of timber.

Check: A crack which separates fibres of wood. It does not extend from one end to other.

Split: When check extend from one end to other.

Twist: When a piece is spirally distorted along its length.

Honey-Combing: Various radial and circular cracks occur in the interior portion of timber due to stress developed while drying.



Preservation of timber



- Natural durable timbers are rarely found. Therefore, ways to improve the durability of cheap timbers are developed.
 - Unprotected timber may decay depending upon the external conditions, species of wood, preliminary conditioning and its structure.
 - Some basic approaches to protect timber are:
 1. Low humidity
 2. Heat
 3. Water insulation
 4. Coating wood surface by paint, polymer films, varnishes etc.
- Preservative treatments do not influence basic properties like Mechanical, Electrical or Chemical properties.

Fire resistance of timber



- Timber is very inflammable
- Heavy timber on burning forms a protective coating of charcoal, which is heat insulator & resists heat penetration.
- The fire resistance of wood can be enhanced either **by impregnating it with effective chemicals.**
 - It include phosphates of ammonia, mixture of ammonium phosphate and ammonium sulphate, borax and boric acid, sodium arsenate & sodium tetra-borate.
- It can also be resisted by **coating** the surface of timber with a layer of **noncombustible paint**
 - Most commonly used flame retardant coatings are cement grouts, clay-sulphate paste; paints such as silicate, chloride, phosphate paints and; emulsions like chloro paraffin